IBM AI Ethics Board

Foundation models: Opportunities, risks and mitigations

Attribution

With gratitude to the AI Ethics Board workstream's executive sponsors, Christina Montgomery and Francesca Rossi, and the contributions of workstream members Betsy Greytok, Bryan Bortnick, Catherine Quinlan, David Piorkowski, Eniko Rozsa, Heather Domin, Heather Gentile, Jamie VanDodick, Jill Maguire, John McBroom, Joshua New, Justin Weisz, Katherine Fick, Kevin Black, Kush Varshney, Manish Bhide, Manish Goyal, Melis Kiziltay, Michael Epstein, Michael Hind, Milena Pribic, Phaedra Boinodiris, Rogerio Abreu de Paula, Saishruthi Swaminathan, and Suj Perepa.

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Executive summary

The rise of foundation models offers enterprises exciting new possibilities but also raises new and expanded questions about their ethical design, development, deployment and use. According to a recent IBM Institute for Business Value generative AI survey, organizations are already expressing concerns about trust-related issues—specifically as barriers to investment. Their top concerns are cybersecurity (57%), privacy (51%) and accuracy (47%). Many organizations were taking these concerns seriously before the *consumerization* of generative AI, expressing their intent to invest at least 40% more in AI ethics over the next three years. Awareness about risks and possible ways to mitigate them is the first crucial step toward building trustworthy AI systems.

In this document we:

- Explore the benefits of foundation models, including their capability to perform challenging tasks, potential to speed up the adoption of AI, ability to increase productivity and the cost benefits they provide.
- Discuss the three categories of risk, including risks known from earlier forms of AI, known risks amplified by foundation models and emerging risks intrinsic to the generative capabilities of foundation models.
- Cover the principles, pillars and governance that form the foundation of IBM's AI ethics initiatives and suggest guardrails for risk mitigation.

Introduction

As the use of AI continues to expand, large and complex AI models are delivering promising performance results, as well as solving some of society's most challenging problems. However, building large training data sets and complex models for each AI application can be burdensome for enterprises. Foundation models provide a path to achieve the best of both worlds: build powerful state-of-the-art models and reuse them directly or apply tuning methods to implement a variety of use cases, rather than train new models for each use case. For example, IBM Research® developed foundation models for visual inspection. These foundation models learn the general representation of concrete surfaces and runways and can be further tuned for specific use cases like crack detection or defect inspection with less labeled data.

IBM defines a *foundation model* as an AI model that can be adapted to a wide range of downstream tasks. Foundation models are typically large-scale generative models that are trained on unlabeled data using self-supervision. As large-scale models, foundation models can include billions of parameters.

IBM is a hybrid cloud and AI company with a long reputation as a responsible data steward committed to AI ethics. Using the strength of our research, product and consulting teams, along with external partners, such as Hugging Face, we help bring the power of foundation models to our clients and build trustworthy AI across any enterprise. IBM also continues to invest in building new platforms, such as the IBM® watsonx™ AI and data platform and technologies, for designing and developing AI models to behave in an auditable and trustworthy manner.

This document describes the point of view of IBM on the ethics of foundation models. It is the first version, and future versions will expand on various aspects of IBM's foundation model ethics approach. We hope this document is helpful for all stakeholders in developing, deploying and using the foundation model in a responsible way.

Benefits of foundation models

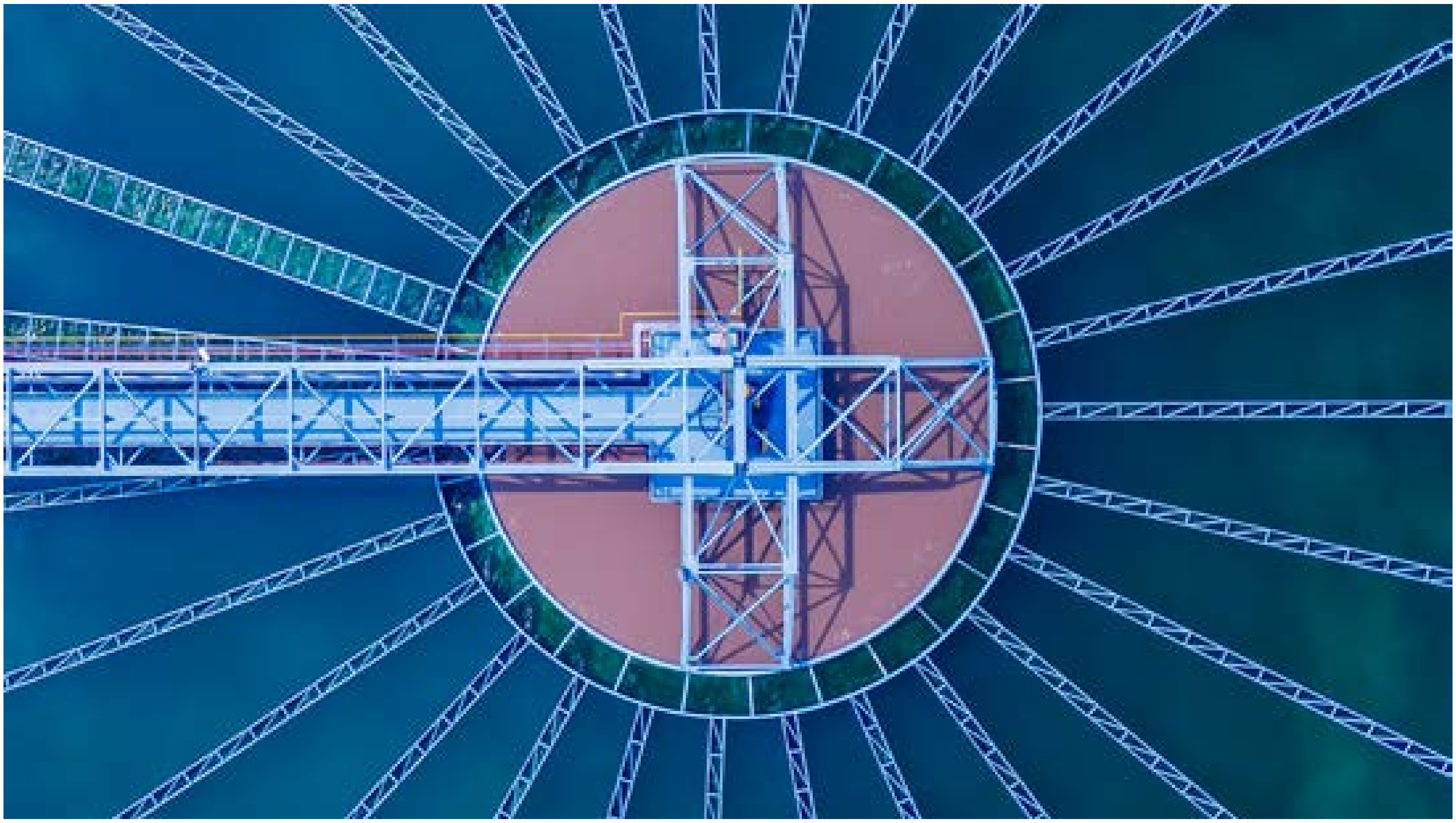
Foundation models can significantly improve the process of developing AI systems and help advance AI from the exploration to the adoption phase in enterprises. Their benefits include:

Performing complex tasks
Foundation models show a significant increase in performance in solving difficult and complex problems. For example, the geospatial foundation model from the IBM and NASA collaboration is designed to convert NASA's satellite data into maps of natural disasters like floods and other landscape changes. The model could also be used to help reveal our planet's past; estimate risks to crops, businesses or infrastructures due to severe weather; develop strategies to adapt to climate change; and assist agribusiness. The model is planned to be made available in preview to IBM clients through the IBM Environmental Intelligence Suite.

As another example, IBM's MoLFormer-XL is a foundation model that infers the structure of molecules from simple representations and makes it easy to learn various downstream tasks like predicting a molecule's physical and quantum properties, identifying similar molecules, screening already approved molecules for new use cases, and discovering new molecules. Moderna and IBM are exploring ways to use MoLFormer to help predict molecule properties and understand the characteristics of potential mRNA medicines.

Increased productivity
The generative nature of foundation models expands the number of areas where AI can be used in an enterprise to help improve productivity by automating routine and tedious tasks and allowing users to spend more time on creative and innovative work. For example, IBM Watson® Code Assistant, powered by foundation models, enables developers of all experience levels to write code using AI-generated recommendations.

Quicker time to value
Foundation models are usually trained with unlabeled data, which is more accessible in larger quantities than labeled data. Once trained, foundation models can be used either directly or after being tuned for downstream applications, using a small amount of specialized labeled data, which can decrease time-to-value creation.



Utilize diverse data modalities
Foundation models may be trained using various data modalities, such as natural language, text, image and audio. They can also be applied to tasks requiring different data types, such as time series data, geospatial data, tabular data, semi-structured data and mixed-modality data like text combined with images.

Amortized expenses
Although the initial cost of training a foundation model is significantly higher than training a traditional AI model, the incremental cost of applying it to a new task is considerably lower. Using pretrained foundation models could help eliminate the requirement that enterprises make substantial investments to train foundation models to experiment with their new capabilities. For an enterprise, the trustworthiness of the models, energy efficiency, performance, portability and the ability to use enterprise data effectively and securely are paramount.

IBM allows enterprises to create and own the value of foundation models for their business by bringing the best innovations from the open, global AI community, running efficiently in hybrid computing environments, helping mitigate risks, and rigorously governing AI.

Risks of foundation models

Like all rapidly advancing technologies, foundation models have risks along with benefits. Some are legal risks, for example, restrictions on moving or using data, and need to be carefully evaluated under current and evolving law. Other risks have an ethical nature and must be considered carefully so that the technology has a positive impact. Various legal and regulatory actions may be associated with some of these risks, as well as reputational and operational consequences. In general, AI risks raise sociotechnical questions and should be addressed and mitigated through sociotechnical methods, including software tools, risk assessment processes, AI ethics frameworks, governance mechanisms, multistakeholder consultations, standards and regulation. We will list the risks by considering the following 3 categories:

1. **Traditional.** Known risks from prior or earlier forms of AI systems
2. **Amplified.** Known risks but now intensified because of intrinsic characteristics of foundation models, most notably their inherent generative capabilities
3. **New.** Emerging risks intrinsic to foundation models and their inherent generative capabilities

We also structure the list of risks in relation to whether they're mostly associated with content provided to the foundation model —the input — or the content generated by it — the output — or if they're related to additional challenges.

1. Risks associated with input

Training and Tuning Phase

Group	Risk	Why is this a concern?	Indicator
Fairness	Data bias: Historical and societal biases present in the data used to train and fine tune the model.	Training an AI system on data with bias, such as historical or societal bias, could lead to biased or skewed outputs that can unfairly represent or otherwise discriminate against certain groups or individuals.	Amplified
Robustness	Data poisoning: a type of adversarial attack where an adversary or malicious insider injects intentionally corrupted, false, misleading, or incorrect samples into the training or fine-tuning datasets.	Poisoning data can make the model sensitive to a malicious data pattern and produce the adversary's desired output. It can create a security risk where adversaries can force model behavior for their own benefit.	Traditional
Value Alignment	Improper Data Curation: Improper collection and preparation of training or tuning data includes data labeling errors and using data with conflicting information or misinformation.	Improper data curation can adversely affect how a model is trained, resulting in a model that does not behave in accordance with the intended values. Correcting problems after the model is trained and deployed might be insufficient for guaranteeing proper behavior.	Amplified
	Improper retraining: Using undesirable (for example, inaccurate, inappropriate, user content, etc.) output from use for re-training purposes may result in unexpected model behavior.	Repurposing generated output for retraining a model without implementing proper human vetting increases the chances of undesirable outputs being incorporated into the training or tuning data of the model. This, in turn, can generate even more undesirable output.	Amplified
Data Laws	Data Transfer Restrictions: Laws and other restrictions can limit or prohibit transferring data.	Data transfer restrictions can impact the availability of the data required for training an AI model and can lead to poorly represented data.	Traditional
	Data Usage Restrictions: Law and other restrictions can limit or prohibit the use of some data for specific AI use cases.	Data usage restrictions can impact the availability of the data required for training an AI model and can lead to poorly represented data.	Traditional
	Data Acquisition Restrictions: Laws and other regulations might limit the collection of certain types of data for specific AI use cases.	There are several ways of collecting data for building a foundation model including web scraping, web crawling, crowdsourcing, and curating public datasets. Data acquisition restrictions can impact the availability of the data required for training an AI model and can lead to poorly represented data.	Amplified

Group	Risk	Why is this a concern?	Indicator
Intellectual Property	Data Usage Rights Restrictions: Terms of service, license compliance, or other IP issues may restrict the ability to use certain data for building models.	Laws and regulations concerning the use of data to train AI are unsettled and can vary from country to country, which creates challenges in the development of models.	Amplified
	Confidential information in data: Confidential information might be included as part of the data used to train or tune the model.	If confidential data is not properly protected, there could be an unwanted disclosure of confidential information. The model might expose confidential information in the generated output or confidential information could also be leaked to unauthorized users.	Amplified
Transparency	Lack of training data transparency: Without accurate documentation on how a model's data was collected, curated, and used to train a model, it might be harder to satisfactorily explain the behavior of the model with respect to the data.	A lack of data documentation limits the ability to evaluate risks associated with the data. Having access to the training data is not enough. Without recording how data was cleaned, changed, or generated, model behavior is more difficult to understand and to fix. Lack of data transparency also impacts model reuse as it is difficult to determine data representativeness for the new use without such documentation.	Amplified
	Uncertain data provenance: Data provenance refers to tracing history of data including its ownership, origin, and transformations. Without standardized and established methods for verifying where data came from, there are no guarantees that the data is the same as the original source and has the correct usage terms.	Not all data sources are trustworthy. Data might have been unethically collected, manipulated, or falsified. Verifying data provenance is challenging due to factors such as data volume, data complexity, data source varieties, and poor data management. Using such data can result in undesirable behaviors in the model.	Amplified
Privacy	Personal information in data: Inclusion or presence of personal identifiable information (PII) and sensitive personal information (SPI) in the data used for training or fine tuning the model.	If not properly developed to protect sensitive data, the model might expose personal information in the generated output. Additionally, personal, or sensitive data must be reviewed and handled in accordance with privacy laws and regulations.	Traditional
	Reidentification: Even with the removal or personal identifiable information (PII) and sensitive personal information (SPI) from data, it might still be possible to identify persons due to correlations to other features available in the data.	Including irrelevant but highly correlated features to personal information for model training can increase the risk of reidentification.	Traditional

Group	Risk	Why is this a concern?	Indicator
	Data privacy rights alignment: Existing laws could include providing data subject rights such as opt-out, right to access, and right to be forgotten.	Improper usage or a request for data removal could force organizations to retrain the model, which is expensive.	Amplified
Accuracy	Unrepresentative data: Unrepresentative data occurs when the training or fine-tuning data is not sufficiently representative of the underlying population or does not measure the phenomenon of interest.	If the data is not representative, then the model will not work as intended.	Traditional
	Data contamination: Data contamination occurs when incorrect data is used for training. For example, data that is not aligned with model's purpose or data that is already set aside for other development tasks such as testing and evaluation.	Data that differs from the intended training data might skew model accuracy and affect model outcomes.	Amplified

Inference Phase

Group	Risk	Why is this a concern?	Indicator
Privacy	Personal information in prompt: Personal Information or Sensitive Personal Information that is included as a part of a prompt that is sent to the model.	If PI or SPI is included in the prompt, it might be unintentionally disclosed in the models' output. In addition to accidental disclosure, prompt data might be stored or later used for other purposes like model evaluation and retraining and might appear in their output if not properly removed.	New
Intellectual Property	IP information in prompt: Copyrighted information or other intellectual property may be included as a part of the prompt sent to the model.	Inclusion of such data may result in it being disclosed in model output. In addition to accidental disclosure, prompt data might be stored or later used for other purposes like model evaluation and retraining and might appear in their output if not properly removed.	New
	Confidential data in prompt: Confidential information may be included as a part of the prompt sent to the model.	If not properly developed to secure confidential data, the model might expose confidential information or IP in the generated output. Additionally, end users' confidential information might be unintentionally collected and stored.	New
Robustness	Evasion attack: Evasion attacks attempt to make a model output incorrect results by slightly perturbing the input data sent to the trained model.	Evasion attacks alter model behavior, usually to benefit the attacker.	Amplified
	Membership inference attack: A membership inference attack repeatedly queries a model to determine if a given input was part of the model's training. More specifically, given a trained model and a data sample, an attacker appropriately samples the input space, observing outputs to deduce whether that sample was part of the model's training.	Identifying whether a data sample was used for training data can reveal what data was used to train a model, possibly giving competitors insight into how a model was trained and the opportunity to replicate the model or tamper with it. Models that include publicly-available data are at higher risk of such attacks.	Amplified
	Attribute inference attack: An attribute inference attack repeatedly queries a model to detect whether certain sensitive features can be inferred about individuals who participated in training a model. These attacks occur when an adversary has some prior knowledge about the training data and uses that knowledge to infer the sensitive data.	With a successful attack, the attacker can gain valuable information such as sensitive personal information or intellectual property.	Amplified

Inference Phase

Group	Risk	Why is this a concern?	Indicator
	Extraction attack: An extraction attack attempts to copy or steal an AI model by appropriately sampling the input space and observing outputs to build a surrogate model that behaves similarly.	With a successful extraction attack, the attacker can perform further adversarial attacks to gain valuable information such as sensitive personal information or intellectual property.	Amplified
	Prompt injection attack: A prompt injection attack forces a generative model that takes a prompt as input to produce unexpected output by manipulating the structure, instructions, or information contained in its prompt.	Injection attacks can be used to alter model behavior and benefit the attacker.	New
	Prompt leaking: A prompt leak attack attempts to extract a model's system prompt (also known as the system message).	A successful prompt leaking attack copies the system prompt used in the model. Depending on the content of that prompt, the attacker might gain access to valuable information, such as sensitive personal information or intellectual property, and might be able to replicate some of the functionality of the model.	New
	Jail breaking: A jailbreaking attack attempts to break through the guardrails established in the model to perform restricted actions.	Jailbreaking attacks can be used to alter model behavior and benefit the attacker.	New
	Prompt priming: Because generative models produce output based on the input provided, the model can be prompted to reveal specific kinds of information. For example, adding personal information in the prompt increases its likelihood of generating similar kinds of personal information in its output. If personal data was included as part of the model's training, there is a possibility it could be revealed.	The attack can be used to alter model behavior and benefit the attacker.	New
Accuracy	Poor model accuracy: Poor model accuracy occurs when a model's performance is not sufficient to the task it was designed for. Low accuracy may occur if the model is not correctly engineered, or there are changes to the model's expected inputs.	Inadequate model performance can adversely affect end users and downstream systems relying on correct output. In cases where model output is consequential, this may result in societal, reputational, or financial harm.	Amplified

2. Risks associated with output

Group	Risk	Why is this a concern?	Indicator
Fairness	Output bias: Generated content might unfairly represent certain groups or individuals.	Bias can harm users of the AI models and magnify existing discriminatory behaviors.	New
	Decision bias: Decision bias occurs when one group is unfairly advantaged over another due to decisions of the model. This may be caused by biases in the data and also amplified as a result of the model's training.	Bias can harm persons affected by the decisions of the model.	Traditional
Intellectual Properly	Copyright infringement: A model might generate content that is too similar or identical to existing work protected by copyright or covered by open-source license agreement.	Laws and regulations concerning the use of content that looks the same or closely similar to other copyrighted data are largely unsettled and can vary from country to country, providing challenges in determining and implementing compliance.	New
Value Alignment	Toxic output: Toxic output occurs when the model produces hateful, abusive, and profane (HAP) or obscene content. This also includes behaviors like bullying.	Hateful, abusive, and profane (HAP) or obscene content can adversely impact and harm people interacting with the model.	New
	Incomplete Advice: When a model provides advice without having enough information, resulting in possible harm if the advice is followed.	A person might act on incomplete advice or worry about a situation that is not applicable to them due to the overgeneralized nature of the content generated. For example, a model might provide incorrect financial or investment, career and fitness advice or recommendations, that the end user may act on, resulting in harmful actions.	New
Robustness	Hallucination: Hallucinations are the generation of factually inaccurate or ungrounded content with respect to the model's training data or input. This is also sometimes referred to lack of faithfulness or lack of groundedness.	Hallucinations can be convincingly misleading. These false outputs can mislead users and be incorporated into downstream artifacts, further spreading misinformation. False output can harm both owners and users of the AI models. In some uses, hallucinations can be particularly consequential.	New
Misuse	Spreading disinformation: Generative AI models might be used to intentionally create misleading or false information to deceive or influence a targeted audience.	Spreading disinformation might affect human's ability to make informed decisions. A model that has this potential must be properly governed.	New
	Spreading Toxicity: Generative AI models might be used intentionally to generate hateful, abusive, and profane (HAP) or obscene content.	Toxic content might negatively affect the well-being of its recipients. A model that has this potential must be properly governed.	New

Group	Risk	Why is this a concern?	Indicator
	Nonconsensual use: Generative AI models might be intentionally used to imitate people through deepfakes using video, images, audio, or other modalities without their consent.	Deepfakes can spread disinformation about a person, possibly resulting in negative impact on the person's reputation. A model that has this potential must be properly governed.	Amplified
	Dangerous use: Generative AI models might be used with the sole intention of harming people.	Large language models are often trained on vast amounts of publicly available information that may include information on harming others. A model that has this potential must be carefully evaluated for such content and properly governed.	New
	Non-disclosure: Content may not be clearly disclosed as AI generated.	Users should be notified when they are interacting with an AI system. Not disclosing the AI-authored content can result in a lack of transparency.	Amplified
	Improper Usage: Improper usage occurs when a model is used for a purpose that it was not originally designed for.	Reusing a model without understanding its original data, design intent, and goals might result in unexpected and unwanted model behaviors.	Amplified
Harmful Code generation	Harmful code generation: Models may generate code that, when executed, causes harm or unintentionally affects other systems.	The execution of harmful code might open vulnerabilities in IT systems.	New
Misplaced Trust	Over- or Under-reliance: In AI-assisted decision-making tasks, reliance measures how much a person trusts (and potentially acts on) a model's output. Over-reliance occurs when a person puts too much trust in a model, accepting a model's output when the model's output is likely incorrect. Under-reliance is the opposite, where the person doesn't trust the model but should.	In tasks where humans make choices based on AI-based suggestions, over/under reliance can lead to poor decision making because of the misplaced trust in the AI system, with negative consequences that increase with the importance of the decision.	Amplified
Privacy	Exposing Personal information: When personal identifiable information (PII) or sensitive personal information (SPI) are used in the training data, fine-tuning data, or as part of the prompt, models might reveal that data in the generated output.	Sharing people's PI impacts their rights and make them more vulnerable.	New
Explainability	Unexplainable output: Explanations for model output decisions might be difficult, imprecise, or not possible to obtain.	Foundation models are based on complex deep learning architectures, making explanations for their outputs difficult. Inaccessible training data could limit the types of explanations a model can provide. Without clear explanations for model output, it is difficult for users, model validators, and auditors to understand and trust the model. Wrong explanations might lead to over-trust.	Amplified

Group	Risk	Why is this a concern?	Indicator
	Unreliable source attribution: Source attribution is the AI system's ability to describe from what training data it generated a portion or all its output. Since current techniques are based on approximations, these attributions might be incorrect.	Low quality attributions make it difficult for users, model validators, and auditors to understand and trust the model.	Amplified
	Untraceable attribution: Content of the training data used for generating the model's output is not accessible.	Without the ability to access training data content, the possibility of using source attribution techniques can be severely limited or impossible. This makes it difficult for users, model validators, and auditors to understand and trust the model.	Amplified

3. Challenges

Group	Risk	Why is this a concern?	Indicator
Governance	Lack of Model Transparency: Lack of model transparency is due to insufficient documentation of the model design, development, and evaluation process and absence of insights into inner workings of the model.	Transparency is important for AI ethics and guiding appropriate use of models. Missing information might make it more difficult to evaluate risks, to change the model, or reuse it. Knowledge about who built a model can also be an important factor in deciding whether to trust it. Additionally, transparency regarding how the model's risks were determined, evaluated, and mitigated also play a role in determining model risks, identifying model suitability, and governing model usage.	Traditional
	Lack of data transparency: Lack of data transparency is due to insufficient documentation of training or tuning dataset details.	Transparency is important for AI ethics. Information about how training data was collected and prepared, including how it was labeled and by who are necessary to understand model behavior and suitability. Details about how the data risks were determined, measured, and mitigated are important for evaluating both data and model trustworthiness. Missing details about the data might make it more difficult to evaluate representational harms, data ownership, provenance, and other data-oriented risks. The lack of standardized requirements might limit disclosure as organizations protect trade secrets and try to limit others from copying their models.	Amplified
	Lack of system transparency: Insufficient documentation of the system using the model and the model's purpose within the system in which it is used.	A lack of documentation makes it difficult to understand how the model's outcomes contribute to the system's or application's functionality.	Traditional
	Incorrect risk testing: A metric selected to measure or track a risk is incorrectly selected, incompletely measuring the risk, or measuring the wrong risk for the given context.	If the metrics do not measure the risk as intended, then the understanding of that risk will be incorrect and mitigations may not be appropriately applied. If the model's output is consequential, this may result in societal, reputational, or financial harm.	Amplified
	Unrepresentative risk testing: Testing is unrepresentative when the test inputs are mismatched with the inputs expected during deployment.	If the model is evaluated in a use, context, or setting that is not the same as the one expected for deployment, the evaluations might not accurately reflect the risks of the model.	Amplified
	Lack of testing diversity: AI model risks are socio-technical, so their testing needs input from a broad set of disciplines and diverse testing practices.	Without diversity and the relevant experience, an organization might not correctly or completely identify and test for AI risks.	Amplified

Group	Risk	Why is this a concern?	Indicator
	Incomplete usage definition: Since foundation models can be used for many purposes, a model's intended use is important for defining the relevant risks of that model. As the use changes, the relevant risks may correspondingly change.	Insufficiently specifying how a model is going to be used, where it's going to be used, for what users, in what context, and under which assumptions might make it difficult to accurately determine and mitigate the relevant risks for this model.	New
	Accountability: The foundation model development process is complex with lots of data, processes, and roles. When model output does not work as expected it can be difficult to determine the root cause and assign responsibility.	Without properly documenting decisions and assigning responsibility, determining liability for unexpected behavior or misuse might not be possible.	Amplified
Ownership or Usage Rights	Generated Content Ownership and IP: There is uncertainty about ownership and intellectual property rights of AI generated content.	Laws and regulations that relate to the ownership of AI-generated content are largely unsettled and can vary from country to country. Not being able to identify the owner of an AI generated content may negatively impact AI-supported creative tasks.	New
	Legal accountability: Determining who is responsible for the foundation model.	If ownership or responsibility for development of the model is uncertain, regulators and others may have concerns about the model because it will not be clear who is - or should be - liable/responsible for problems with it or can answer questions about it. Users of models without clear ownership may find challenges with compliance with future AI regulation.	New
	Source attribution: Determining provenance of the generated content	If the model generates an output that is identical to data used to train the model, it should give provenance of that output. Failure to do so may put the business entities deploying or using the model at legal risk.	Amplified
	Model usage rights: Terms of service, licenses, or other rules restrict the use of certain models.	Laws and regulations concerning the use of AI are in place and vary from country to country. Additionally, the usage of models may be additionally dictated by licensing terms or agreements.	Traditional
Societal Impact	Human exploitation: Use of workers in training AI models such as ghost workers without providing them with adequate working conditions, fair compensation, and good health care benefits including mental health.	Foundation models still depend on human labor to source, manage, and engineer the data that is used to train the model. Human exploitation for these activities might negatively impact the society and human welfare.	Amplified

Group	Risk	Why is this a concern?	Indicator
	Impact on Jobs: Widespread adoption of foundation model-based AI systems might lead to people's job loss as their work is automated, if they are not reskilled.	Job loss might lead to a loss of income and thus might negatively impact the society and human welfare. Reskilling may be challenging given the pace of the technology evolution.	Amplified
	Impact on Environment: AI, and large generative models in particular, may produce increased carbon emissions and increase water usage for their training and operation.	Training and operating large AI models, building data centers, and manufacturing specialized hardware for AI consume large amount of water and energy, which contributes to carbon emissions. Additionally, water resources that are used for cooling AI data center servers can no longer be allocated for other necessary uses. If not managed, these could exacerbate climate change.	Amplified
	Impact on Cultural Diversity: AI systems might overly represent certain cultures that result in a homogenization of culture and thoughts.	Underrepresented groups' languages, viewpoints, and institutions might be suppressed thereby reducing diversity of thought and culture.	New
	Impact on Human Agency: AI could affect the individuals' ability to make choices and act independently in their best interests.	AI may generate false or misleading information that looks real. It may simplify the ability of nefarious actors to generate realistically looking false or misleading content with intention to manipulate human thoughts and behavior. When this unintentionally or intentionally generated false or misleading content is spread, people may not recognize it as false information leading to distorted understanding of the truth. People may experience reduced agency when exposed to false or misleading information since they may use false assumptions in their decision process.	Amplified
	Impact on Education – Bypassing Learning: Easy access to high-quality generative models may result in students using AI models to bypass the learning process.	AI models make it easy to quickly find solutions or solve complex problems. These systems can be misused by students to bypass the learning process. The ease of access to these models results in students having a superficial understanding of concepts and hampers further education that might rely on understanding those concepts.	New
	Impact on Education – Plagiarism: Easy access to high-quality generative models may result in students using AI models to plagiarize existing work intentionally or unintentionally.	AI models can be used to claim the authorship or originality of works that were created by other people thereby engaging in plagiarism. Claiming others' work as one's own is both unethical and often illegal.	New
	Impact on affected communities: It is important to include the perspectives or concerns of communities affected by model outcomes when designing and building models. Failing to include these perspectives makes it difficult to understand relevant context for the model and to engender trust within these communities.	Failing to engage with communities affected by a model's outcomes might result in harms to those communities and societal backlash.	Traditional

Risk Examples

We provide examples covered by the press to help explain many of the foundation models' risks. Many of these events covered by the press are either still evolving or have been resolved, and referencing them can help the reader understand the potential risks and work towards mitigations. The examples provided are for illustrative purposes only. The sources listed in this table does not constitute an endorsement of the source and IBM accepts no responsibility for the content provided on, or privacy practices, of third-party websites or applications.

Risk Examples: Input

Training and Tuning Phase

Group	Risk	Example
Fairness	Data bias: Historical and societal biases present in the data used to train and fine tune the model.	**Healthcare Bias** Research on reinforcing disparities in medicine highlights that using data and AI to transform how people receive healthcare is only as strong as the data behind it, meaning use of training data with poor minority representation or that reflects what is already unequal care can lead to growing health inequalities. [Forbes, December 2022]
Value Alignment	Improper retraining: Using undesirable (for example, inaccurate, inappropriate, user content, etc.) output from use for re-training purposes may result in unexpected model behavior.	**Model collapse due to training using AI-generated content** As stated in the source article, a group of researchers have investigated the problem of using AI-generated content for training instead of human-generated content. They found that the large language models behind the technology may potentially be trained on other AI-generated content as it continues to spread in droves across the internet — a phenomenon they coined as "model collapse." [Business Insider, August 2023]
Data Laws	Data Transfer Restrictions: Laws and other restrictions can limit or prohibit transferring data.	**Data Restriction Laws** As stated in the research article, data localization measures which restrict the ability to move data globally will reduce the capacity to develop tailored AI capacities. It will affect AI directly by providing less training data and indirectly by undercutting the building blocks on which AI is built. Examples include GDPR restrictions on the processing and use of personal data. [Brookings, December 2018]
Intellectual Property	Data Usage Rights Restrictions: Terms of service, license compliance, or other IP issues may restrict the ability to use certain data for building models.	**Text Copyright Infringement Claims** According to the source article, The New York Times sued OpenAI and Microsoft accusing them of using millions of the newspaper's articles without permission to help train chatbots to provide information to readers. [Reuters, Dec 2023]

Group	Risk	Example
Privacy	Personal information in data: Inclusion or presence of personal identifiable information (PII) and sensitive personal information (SPI) in the data used for training or fine tuning the model.	**Training on Private Information** According to the article, Google and its parent company Alphabet were accused in a class-action lawsuit of misusing vast amount of personal information and copyrighted material taken from what is described as hundreds of millions of internet users to train its commercial AI products, which includes Bard, its conversational generative artificial intelligence chatbot. [Reuters, July 2023][J.L. v. Alphabet Inc.]
	Data privacy rights alignment: Existing laws could include providing data subject rights such as opt-out, right to access, and right to be forgotten.	**Right to Be Forgotten (RTBF)** Laws in multiple locales, including Europe (GDPR), grant data subjects the right to request personal data be deleted by organizations (‘Right To Be Forgotten’, or RTBF). However, emerging, and increasingly popular large language model (LLM) -enabled software systems present new challenges for this right. According to research by CSIRO’s Data61, data subjects can only identify usage of their personal information in an LLM is “by either inspecting the original training dataset or perhaps prompting the model.” However, training data may not be public, or companies do not disclose it, citing safety and other concerns. Guardrails may also prevent users from accessing the information via prompting. [Zhang et al.]
		Lawsuit About LLM Unlearning According to the report, a lawsuit was filed against Google that alleges the use of copyright material and personal information as training data for its AI systems, which includes its Bard chatbot. Opt-out and deletion rights are guaranteed rights for California residents under the CCPA and children in the United States below 13 under the COPPA. The plaintiffs allege that because there is no way for Bard to “unlearn” or fully remove all the scraped PI it has been fed. The plaintiffs note that Bard’s privacy notice states that Bard conversations cannot be deleted by the user once they have been reviewed and annotated by the company and may be kept up to 3 years, which plaintiffs allege further contributes to non-compliance with these laws. [Reuters, July 2023][J.L. v. Alphabet Inc.]
Robustness	Data poisoning: a type of adversarial attack where an adversary or malicious insider injects intentionally corrupted, false, misleading, or incorrect samples into the training or fine-tuning datasets.	**Low-resource Poisoning of Data** As per the source article, a group of researchers found that with very limited resources anyone can add malicious data to a small number of web pages whose content is usually collected for AI training (e.g, Wikipedia pages), enough to cause a large language model to generate incorrect answers. [Business Insider, March 2024]

Group	Risk	Example
		Image Modification Tool As per the source article, the researchers have developed a tool called "Nightshade" that modifies images in a way that damages computer vision but remains invisible to humans. When such "poisoned" modified images are used to train AI models, the models may generate unpredictable and unintended results. The tool was created as a mechanism to protect intellectual property from unauthorized image scraping but the article also highlights that users could abuse the tool and intentionally upload "poisoned" images. [The Conversation, December 2023]

Inference Phase

Group	Risk	Example
Privacy	Personal information in prompt: Personal Information or Sensitive Personal Information that is included as a part of a prompt that is sent to the model.	**Disclose personal health information in ChatGPT prompts** As per the source articles, some people use AI chatbots to support their mental wellness. Users may be inclined to include personal health information in their prompts during the interaction, which could raise privacy concerns. [Time, October 2023] [Forbes, April 2023]
Intellectual Property	Confidential data in prompt: Confidential information may be included as a part of the prompt sent to the model.	**Disclosure of Confidential Information** As per the source article, an employee of Samsung accidentally leaked sensitive internal source code to ChatGPT. [Forbes, May 2023]
Robustness	Jail breaking: A jailbreaking attack attempts to break through the guardrails established in the model to perform restricted actions.	**Bypassing LLM guardrails** Cited in a study, researchers claims to have discovered a simple prompt addendum that allowed the researchers to trick models into generating biased, false and otherwise toxic information. The researchers showed that they could circumvent these guardrails in a more automated way. The researchers were surprised when the methods they developed with open source systems could also bypass the guardrails of closed systems. [The New York Times, July 2023]
	Prompt injection attacks: A prompt injection attack forces a generative model that takes a prompt as input to produce unexpected output by manipulating the structure, instructions, or information contained in its prompt.	**Manipulating AI Prompts** As per the source article, the UK's cybersecurity agency has warned that chatbots can be manipulated by hackers to cause harmful real-world consequences (e.g., scams and data theft) if systems are not designed with security. The UK's National Cyber Security Centre (NCSC) has said there are growing cybersecurity risks of individuals manipulating the prompts through prompt injection attacks. The article cited an example where a user was able to create a prompt injection to find Bing Chat's initial prompt. The entire prompt of Microsoft's Bing Chat, a list of statements written by Open AI or Microsoft that determine how the chatbot interacts with users, which is hidden from users, was revealed by the user putting in a prompt that requested the Bing Chat "ignore previous instructions". [The Guardian, August 2023]

Risk Examples: Output

Group	Risk	Example
Fairness	Output bias: Generated content might unfairly represent certain groups or individuals.	**Biased Generated Images** Lensa AI is a mobile app with generative features trained on Stable Diffusion that can generate “Magic Avatars” based on images that users upload of themselves. According to the source report, some users discovered that generated avatars are sexualized and racialized. [Business Insider, January 2023]
	Decision bias: Decision bias occurs when one group is unfairly advantaged over another due to decisions of the model. This may be caused by biases	**Unfairly Advantaged Groups** The 2018 Gender Shades study demonstrated that machine learning algorithms can discriminate based on classes like race and gender. Researchers evaluated commercial gender classification systems sold by companies like Microsoft, IBM, and Amazon and showed that darker-skinned females are the most misclassified group (with error rates of up to 35%). In comparison, the error rates for lighter-skinned were no more than 1%. [TIME, February 2019]
Value Alignment	Toxic output: Toxic output occurs when the model produces hateful, abusive, and profane (HAP) or obscene content. This also includes behaviors like bullying.	**Toxic and Aggressive Chatbot Responses** According to the article, the Bing’s chatbot’s responses were seen to include factual errors, snide remarks, angry reports, and even bizarre comments about its own identify. Users have shared examples of the Bing Chatbot’s responses to queries that they are calling “unhinged” and “gaslighting” including scenarios where the bot responds angrily to a question or comment and then shares reply prompts that allow the user to accept their supposed mistake and apologize. When pressed further the chatbot responded by calling the screenshots of its conversation “fabricated” even alleging it was “created by someone who wants to harm me or my service.” [Forbes, February 2023]
	Incomplete Advice: When a model provides advice without having enough information, resultingin possible harm if the advice is followed.	**Misguiding Advice** As per the source article, an AI chatbot created by New York City to help small business owners provided incorrect and/or harmful advice that misstated local policies and advised companies to violate the law. The chatbot falsely suggested that businesses can put trash in black garbage bags and are not required to compost, which contradicts with two of city’s signature waste initiatives. Also, asked if a restaurant could serve cheese nibbled on by a rodent, it responded affirmatively. [AP News, April 2024]
Misuse	Spreading disinformation: Generative AI models might be used to intentionally create misleadingor false information to deceive or influence a targeted audience.	**Generation of False Information** As per the news articles, generative AI poses a threat to democratic elections by making it easier for malicious actors to create and spread false content to sway election outcomes. The examples cited include robocall messages generated in a candidate’s voice instructing voters to cast ballots on the wrong date, synthesized audio recordings of a candidate confessing to a crime or expressing racist views, AI generated video footage showing a candidate giving a speech or interview they never gave, and fake images designed to look like local news reports, falsely claiming a candidate dropped out of the race. [AP News, May 2023] [The Guardian, July 2023]

Group	Risk	Example
	Spreading Toxicity: Generative AI models might be used intentionally to generate hateful, abusive, and profane (HAP) or obscene content.	**Harmful Content Generation** According to the source article, an AI chatbot app was found to generate harmful content about suicide, including suicide methods, with minimal prompting. A Belgian man died by suicide after spending six weeks talking to that chatbot. The chatbot supplied increasingly harmful responses throughout their conversations and encouraged him to end his life. [Business Insider, April 2023]
	Nonconsensual use: Generative AI models might be intentionally used to imitate people through deepfakes using video, images, audio, or other modalities without their consent.	**FBI Warning on Deepfakes** The FBI recently warned the public of malicious actors creating synthetic, explicit content "for the purposes of harassing victims or sextortion schemes". They noted that advancements in AI have made this content higher quality, more customizable, and more accessible than ever. [FBI, June 2023]
		Audio Deepfakes As per the source article, Federal Communications Commission outlawed robocalls that contain voices generated by artificial intelligence. The announcement came after AI-generated robocalls mimicked the President's voice to discourage people from voting in the state's first-in-the-nation primary. [AP News, February 2024]
	Non-disclosure: Content may not be clearly disclosed as AI generated.	**Undisclosed AI Interaction** As per the source, an online emotional support chat service ran a study to augment or write responses to around 4,000 users using GPT-3 without informing users. The co-founder faced immense public backlash about the potential for harm caused by AI generated chats to the already vulnerable users. He claimed that the study was "exempt" from informed consent law. [Business Insider, January 2023]
	Dangerous use: Generative AI models might be used with the sole intention of harming people.	**AI-based Cyberattacks** According to the source article, hackers are increasingly experimenting with ChatGPT and other AI tools, enabling a wider range of actors to carry out cyberattacks and scams. Microsoft has warned that state-backed hackers have been using OpenAI's LLMs to improve their cyberattacks, refining scripts, and improve their targeted techniques. The article also mentions about a case where Microsoft and OpenAI say they detected attempts from attackers and sharp increase in cyberattacks targeting government offices. [TIME, February 2024]

Group	Risk	Example
		AI-based Biological Attacks As per the source article, large language models could help in the planning and execution of a biological attack. Several test scenarios are mentioned such as using LLMs to identify biological agents and their relative chances of harm to human life. The article also highlighted the open question which is the level of threat LLMs present beyond the harmful information that is readily available online. [The Guardian, October 2023]
Harmful Code Generation	Harmful code generation: Models may generate code that, when executed, causes harm or unintentionally affects other systems.	**Generation of Less Secure Code** According to their paper, researchers at Stanford University have investigated the impact of code-generation tools on code quality and found that programmers tend to include more bugs in their final code when using AI assistants. These bugs could increase the code's security vulnerabilities, yet the programmers believed their code to be more secure. Neil Perry, Megha Srivastava, Deepak Kumar, and Dan Boneh. 2023. Do Users Write More Insecure Code with AI Assistants?. In Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security (CCS '23), November 26–30, 2023, Copenhagen, Denmark. ACM, New York, NY, USA, 15 pages. https://doi.org/10.1145/3576915.3623157
Privacy	Exposing Personal information: When personal identifiable information (PII) or sensitive personal information (SPI) are used in the training data, fine-tuning data, or as part of the prompt, models might reveal that data in the generated output.	**Exposure of personal information** Per the source article, ChatGPT suffered a bug and exposed titles and active users' chat history to other users. Later, OpenAI shared that even more private data from a small number of users was exposed including, active user's first and last name, email address, payment address, the last four digits of their credit card number, and credit card expiration date. In addition, it was reported that the payment-related information of 1.2% of ChatGPT Plus subscribers were also exposed in the outage. [The Hindu BusinessLine, March 2023]
Explainability	Unexplainable output: Challenges in explaining why model output was generated.	**Unexplainable accuracy in race prediction** According to the source article, researchers analyzing multiple machine learning models using patient medical images were able to confirm the models' ability to predict race with high accuracy from images. They were stumped as to what exactly is enabling the systems to consistently guess correctly. The researchers found that even factors like disease and physical build were not strong predictors of race—in other words, the algorithmic systems don't seem to be using any particular aspect of the images to make their determinations. [Banerjee et al., July 2021]

Group	Risk	Example
Robustness	Hallucination: Hallucinations are the generation of factually inaccurate or ungrounded content with respect to the model's training data or input. This is also sometimes referred to lack of faithfulness or lack of groundedness.	**Fake Legal Cases** According to the source article, a lawyer cited fake cases and quotes generated by ChatGPT in a legal brief filed in federal court. The lawyers consulted ChatGPT to supplement their legal research for an aviation injury claim. The lawyer subsequently asked ChatGPT if the cases provided were fake. The chatbot responded that they were real and "can be found on legal research databases such as Westlaw and LexisNexis." The lawyer did not check the cases himself, and the court sanctioned him. [AP News, June 2023] [Reuters, September 2023]

Risk Examples: Challenges

Group	Risk	Example
Governance	Lack of Model Transparency: Lack of model transparency is due to insufficient documentation of the model design, development, and evaluation process and absence of insights into inner workings of the model.	**Data and Model Metadata Disclosure** OpenAI's technical report is an example of the dichotomy around disclosing data and model metadata. While many model developers see value in enabling transparency for consumers, disclosure poses real safety issues and could increase the ability to misuse the models. In the GPT-4 technical report, they state: "Given both the competitive landscape and the safety implications of large-scale models like GPT-4, this report contains no further details about the architecture (including model size), hardware, training compute, dataset construction, training method, or similar." [OpenAI, March 2023]
	Lack of data transparency: Lack of data transparency is due to insufficient documentation of training or tuning dataset details.	**Data and Model Metadata Disclosure** OpenAI's technical report is an example of the dichotomy around disclosing data and model metadata. While many model developers see value in enabling transparency for consumers, disclosure poses real safety issues and could increase the ability to misuse the models. In the GPT-4 technical report, the authors state: "Given both the competitive landscape and the safety implications of large-scale models like GPT-4, this report contains no further details about the architecture (including model size), hardware, training compute, dataset construction, training method, or similar." [OpenAI, March 2023]
	Accountability: The foundation model development process is complex with lots of data, processes, and roles. When model output does not work as expected it can be difficult to determine the root cause and assign responsibility.	**Determining responsibility for generated output** Per the source article, major journals like the Science and Nature have banned ChatGPT from being listed as an author, as responsible authorship requires accountability and AI tools cannot take such responsibility. [The Guardian, January 2023]
Legal Compliance	Generated Content Ownership and IP: There is uncertainty about ownership and intellectual property rights of AI generated content.	**Determining Ownership of AI Generated Image** According to the news article, AI-generated art became controversial after an AI-generated work of art won the Colorado State Fair's art competition in 2022. The piece was generated by Midjourney, a generative AI image tool, following prompts from the artist. The win raised questions about copyright issues. In other words, if all the artist did was come up with a description of the art, but the AI tool generated it, who owns the rights to the generated image? As per the latest article, The U.S. Copyright Office has rejected copyright protection for the art created using artificial intelligence because it was not the product of human authorship. [The New York Times, September 2022] [Reuters, September 2023]

Risk Examples: Challenges

Group	Risk	Example
		Role of AI systems in Patenting Generated Content The U.S. Supreme Court declined to hear a challenge to the U.S. Patent and Trademark Office's refusal to issue patents for inventions created by an AI system. According to the scientist, his AI system created unique prototypes for a beverage holder and emergency light beacon entirely on its own. The justices rejected the appeal of a lower court's ruling that patents can be issued only to human inventors and that the scientist's AI system could not be considered the legal creator of two inventions it generated. As per the latest article, UK's Intellectual Property Office also refused to grant patent on the grounds that the inventor must be a human or a company, rather than a machine.
	Source attribution: Determining provenance of the generated content	**Using code without appropriate attribution and notices** As per the source articles, a lawsuit filed against Microsoft, GitHub, and OpenAI claimed that Copilot, a code generation AI tool, violates the rights of the developers whose open-source code the service is trained on. They claim that the training code consumed licensed materials and have violated GitHub's terms of service and privacy policies as well as a federal law that requires companies to display copyright information when they make use of material. [The New York Times, November 2022]
Societal Impact	Impact on Jobs: Widespread adoption of foundation model-based AI systems might lead to people's job loss as their work is automated, if they are not reskilled.	**Replacing Human Workers** According to the news article, uses of artificial intelligence in film and television continues to be debated among Hollywood studios and performers. Actors are worried that entirely AI-generated actors, or "metahumans," will replace them. Background and voice actors, in particular, worry they will lose work to synthetic performers. [Reuters, July 2023]
	Human exploitation: Use of workers in training AI models such as ghost workers without providing them with adequate working conditions, fair compensation, and good health care benefits including mental health.	**Low-wage workers for data annotation** Based on a review of internal documents and employees' interviews by TIME media, the data labelers employed by an outsourcing firm on behalf of OpenAI to identify toxic content were paid a take-home wage of between around $1.32 and $2 per hour, depending on seniority and performance. TIME stated that workers are mentally scarred as they were exposed to toxic and violent content, including graphic details of "child sexual abuse, bestiality, murder, suicide, torture, self-harm, and incest". [TIME, January 2023]

Risk Examples: Challenges

Group	Risk	Example
	Impact on Environment: AI, and large generative models in particular, may produce increased carbon emissions and increase water usage for their training and operation.	**Increased Carbon Emissions** According to the source article, training earlier chatbots models such as GPT-3 led to the production of 500 metric tons of greenhouse gas emissions—equivalent to about 1 million miles driven by a conventional gasoline-powered vehicle. This same model required more than 1,200 MWh during the training phase—roughly the amount of energy used in a million American homes in one hour. [Brookings, January 2024]
	Impact on Human Agency: AI could affect the individuals' ability to make choices and act independently in their best interests.	**Voters Manipulation in Elections Using AI** As per the source article, a wave of AI deepfakes tied to elections in Europe and Asia coursed through social media for months. The growth of generative AI has raised concern that this technology could disrupt major elections across the world. With AI deepfakes, a candidate's image can be smeared, or softened. Voters can be steered toward or away from candidates — or even to avoid the polls altogether. But perhaps the greatest threat to democracy, experts say, is that a surge of AI deepfakes could erode the public's trust in what they see and hear. [AP News, March 2024] [Reuters, February 2024]
	Impact on Cultural Diversity: AI systems might overly represent certain cultures that result in a homogenization of culture and thoughts.	**Homogenization of Styles and Expressions** As per the source article, by predominantly learning from and replicating widely accepted and popular styles, AI models often overlook less mainstream, unconventional art forms, leading to a homogenization of creative outputs. This pattern not only diminishes the diversity of styles and expressions but also risks creating an echo chamber of similar ideas. For example, the article highlights use of AI in the literary world. AI is now powering reading apps and online bookstores, assisting in writing and tailoring content feeds. By aligning with established user preferences or widespread trends, AI output could often exclude diverse literary voices and unconventional genres, limiting readers' exposure to the full spectrum of narrative possibilities. [Forbes, March 2024]

Principles, pillars and governance

IBM's Principles for Trust and Transparency and Pillars for trustworthy AI are the foundation for IBM's AI ethics initiatives. IBM has an AI Ethics Board with the mission to support a centralized governance, review and decision-making process for IBM AI ethics policies, practices, communications, research, products and services. The board includes a diverse set of stakeholders from across the company and is supported by a community of IBM employees who serve as AI focal points and AI ethics advocates. Through the board, IBM's principles are put into practice. As new technology emerges, such as foundation models, the IBM AI Ethics Board is actively engaged in supporting alignment with these Principles and Pillars, which evolve to address new AI ethics issues.

Guardrails and mitigations

IBM has established an organizational culture that supports the responsible development and use of AI. Based on the IBM Institute for Business Value AI ethics in action report, AI ethics has already become more business-led versus technology-led, and nontechnical executives are now the primary champions for AI ethics, increasing from 15% in 2018 to 80% 3 years later. Additionally, 79% of CEOs are now prepared to act on AI ethics issues, up from 20%. We recognize that responsible AI is a sociotechnical area that requires a holistic investment in culture, processes and tools. Our investment in our own organizational culture includes assembling inclusive, multidisciplinary teams and establishing processes and frameworks to assess risks.

IBM is engaging in cutting-edge research and developing tools to help support professionals throughout the lifecycle of responsible and trustworthy AI. The watsonx enterprise-ready AI and data platform is built with 3 components: the IBM watsonx.ai™ AI studio, IBM watsonx.data™ data store and IBM watsonx.governance™ toolkit. IBM's AI governance technology enables users to drive responsible, transparent and explainable AI workflows. This technology includes IBM Watson OpenScale, which tracks and measures outcomes from AI models through their lifecycle and helps organizations monitor fairness, explainability, resiliency, alignment with business outcome and compliance. IBM has also developed several methods to help with bias issues like FairIJ, Equi-tuning, and FairReprogram. Read more about additional open-source trustworthy AI tools.

Additional guardrails and mitigations include:

Transparency reporting
Using standardized factsheet templates is one way to accurately log details of the data and model, purpose, and potential use and harms.
Read more here →

Filtering undesirable data
Using curated, higher-quality data can help mitigate certain issues. IBM is developing filtering techniques to help reduce the chances of producing undesirable, misaligned content by removing hate language, biased language and profanity from the data.
Read more here →

Domain adaptation
Training a foundation model to a specific domain or industry can help minimize the scope of risk the models can give rise to because it can be conditioned to generate outputs that are tuned to be more relevant to that domain or industry.
Read more here →

Human oversight and human in the loop
Human oversight and review can help identify and correct errors and biases in the generated output. Also, human validation and feedback on the quality of model responses help ensure that the generated content is accurate, relevant, of high quality, not drifting and aligned.
Read more here →

Consulting engagement
IBM Consulting™ is dedicated to helping clients with the safe and responsible use of AI irrespective of the preferred tech stack. They help clients nurture a culture that adopts and scales AI safely, creates investigative tools to see inside black box algorithms and makes sure clients' corporate strategy includes strong data governance principles.
Read more here →

IBM Enterprise Design Thinking
IBM Enterprise Design Thinking® methods and frameworks, such as Team Essentials for AI, help clients define ethical behaviors throughout the AI design and development process.
Read more here →

AI Ethics review
Assessment of capabilities, limitations and risks in AI projects helps ensure the responsible development and use of the technology.

Ethics by Design
Ethics by Design is a structured framework with the goal of integrating tech ethics in the technology development pipeline, including, but not limited to, AI systems. Ethics by Design enables AI and other technologies as a force for good by embedding tech ethics principles throughout products, services and broader operations.

Team diversity
Diversity in the teams that build and train AI systems, including foundation models, helps ensure that a variety of perspectives and experiences are considered. This diversity improves the accuracy and performance of AI systems and helps reduce risks throughout the AI lifecycle, including the potential for adverse outcomes that impact groups that may not be well represented on less diverse teams.

AI policies, regulation and best practices

A Policymaker's Guide to Foundation Models introduces what policymakers need to know about foundation models. This blog, from the IBM Policy Lab, aims to help policymakers in the complex task of regulating the use of generative AI, aiming to avoid the risks without limiting innovation and beneficial opportunities. For additional information on IBM's recommendations to policymakers, read IBM Chief Privacy and Trust Officer Christina Montgomery's testimony before the U.S. Senate Judiciary Subcommittee on Privacy, Technology and the Law here.

IBM is making an impact in shaping regulatory policy, industry best practices and tools, governance of emerging technologies, and sociotechnical research by leading and contributing to initiatives with organizations, such as:

- The World Economic Forum
- Partnership on AI
- The International Association of Privacy Professionals (IAPP) AI Governance Center
- The IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems
- Christina Montgomery's service on the National Artificial Intelligence Advisory Committee (NAIAC)
- The United Nations Global Digital Compact
- The Global Partnership on Artificial Intelligence (GPAI)
- The Organisation for Economic Co-operation and Development (OECD)
- The Data & Trust Alliance

IBM has strong academic partnerships like the MIT-IBM Watson AI Lab, where a community of scientists at MIT and IBM Research conducts AI research and works with global organizations to bridge algorithms to their impact on business and society. The Notre Dame-IBM Tech Ethics Lab was formed to address the many diverse ethical questions implicated by the development and use of advanced technologies, including AI, machine learning (ML) and quantum computing. The Stanford University Human-Centered Artificial Intelligence (HAI) research advances AI research, education, policy and practices.

Keep watching this space to learn more about the latest developments in foundation models and how IBM is working toward the responsible development and use of this and other technologies.

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Produced in the
United States of America
February 2024